SPACE MISSION

ANALYTICAL DASHBOARDS

End-to-End Data Pipeline · MySQL · Power BI · 5-Page Dashboard

1,584	7	68	333	47
Missions	Countries	Years	Rockets	Launch Sites

Dataset: 1,584 Records | 1957–2025 | 23 Columns | 36→23 Cleaned | 7 SQL Views

PROJECT INTRODUCTION

What is This Project About?

Objective	Analyse 68 years of global space missions to uncover patterns in launch activity, mission success rates, rocket reliability, and orbital strategy across 7 major space programs.
Tools Used	Python (data cleaning) · MySQL (import, type conversion, SQL views) · Power BI Desktop (5-page interactive dashboard)
Dataset	1,584 missions spanning 1957–2025. Original 36-column raw CSV cleaned to 23 final columns with 2 new derived fields (Decade, Spacecraft_Status).
Scope	7 country groups: USA, Russia/USSR, China, India, Europe/ESA, Japan, Other Countries. Covers all mission types from Military/Recon to Human Spaceflight.
Deliverable	Interactive 6-page Power BI report (Home + 5 analysis pages) with 7 SQL views, 7 DAX measures, and synced global slicers.

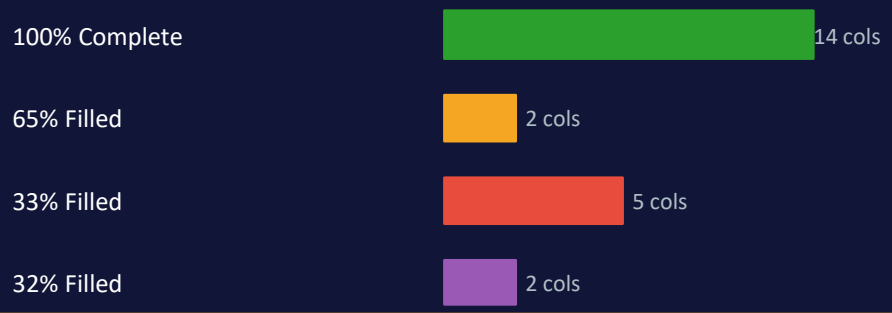
DATASET OVERVIEW

23 Final Columns — By Group

IDENTITY (4)	Mission_Name · Launch_Date · Launch_Year · Decade★
GEOGRAPHY (3)	Country_Group · Country · Agency
VEHICLE (4)	Launch_Vehicle · Rocket_Family · Launch_Site · Launch_Complex
MISSION (2)	Mission_Type (12 cats) · Orbit_Class (13 cats)
PHYSICAL (1)	Satellite_Mass_kg (32% filled — 512 rows)
OUTCOME (2)	Mission_Outcome (4 groups) · Spacecraft_Status★
ECONOMICS (3)	Estimated_Cost_USD_M · Fuel_Type · Operational_Life_Yrs
METADATA (4)	Notes · Launch_ID · Data_Source · Dataset_Source

★ = newly derived column

Column Fill Rates



Cleaning Summary

- 36 raw columns → 23 final columns
- 13 columns dropped (redundant / unusable)
- 7 columns normalised (types, categories)
- 2 new columns derived (Decade, Spacecraft_Status)
- 7 problem date rows fixed manually
- Empty strings converted to NULL (launch_site)
- Cost ranges converted to numeric midpoints

DATA CLEANING PROCESS

1 — Raw Import

Create missions_raw table with ALL columns as VARCHAR — guarantees zero import errors on any data format.

2 — CSV Import

Import MASTER_CLEANED_FINAL.csv via MySQL Workbench Table Data Import Wizard into missions_raw.

3 — Fix 7 Dates

Manually fix 7 problem rows (range dates, planned dates, mixed formats) by mission name using MAKEDATE().

4 — Type Convert

Add clean columns, convert VARCHAR→DATE, SMALLINT, DECIMAL using STR_TO_DATE, CAST, NULLIF, TRIM.

5 — Build Clean Table

CREATE missions_clean with proper data types. INSERT SELECT from missions_raw using converted columns.

Critical Concept — NULL vs Empty String vs 'Unknown'

State	Meaning	Correct SQL Filter	Example Column
NULL	No value — blank cell in CSV	WHERE col IS NOT NULL	satellite_mass_kg
Empty String ""	Cell existed with no content	WHERE col != ""	fuel_type
"Unknown"	Real text value = Unknown	WHERE col NOT IN ("Unknown")	orbit_class

DATA UNDERSTANDING — KEY FIELDS & STRUCTURE

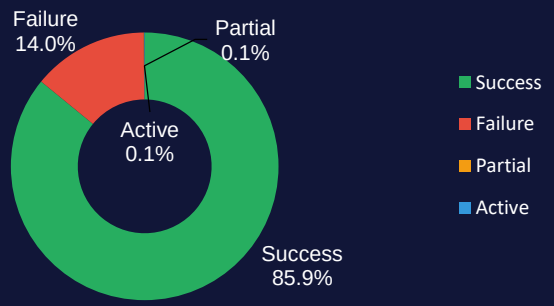
Important Fields Explained

Country_Group	7 clean groups — primary slicer for all dashboards. Never use raw Country column in visuals.
Mission_Outcome	4 values: Success (86%), Failure (14%), Partial, Active. BepiColombo = Active (launched 2018).
Orbit_Class	13 types. Has 'Unknown' and 'Other' text — filter: NOT IN ('Unknown','Other').
Fuel_Type	5 families. Missing stored as empty string '' not NULL — filter: fuel_type != ''.
Decade	Derived column: '1960s','1970s'… for decade-level grouping. 100% filled.
Spacecraft_Status	Derived: Decommissioned / Active / Lost-Failed / Planned. Only 33% filled.
Estimated_Cost_USD_M	Numeric midpoints from ranges. Only 503/1584 rows (32%) — always add coverage note.

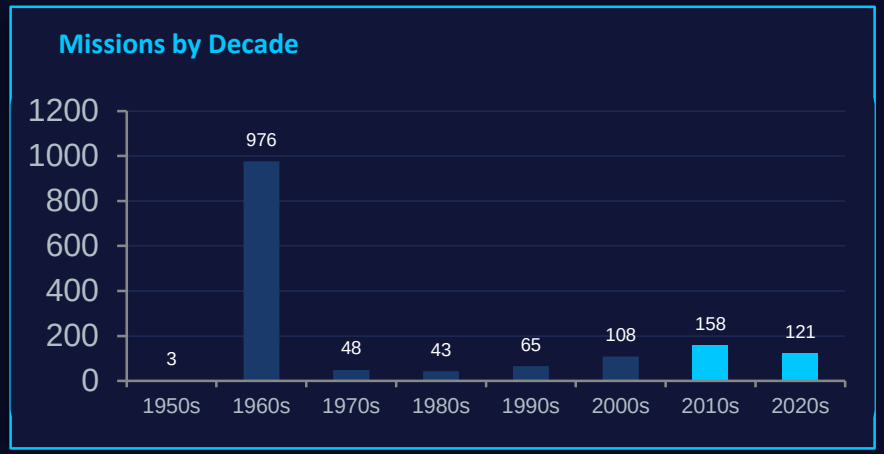
Data Pipeline

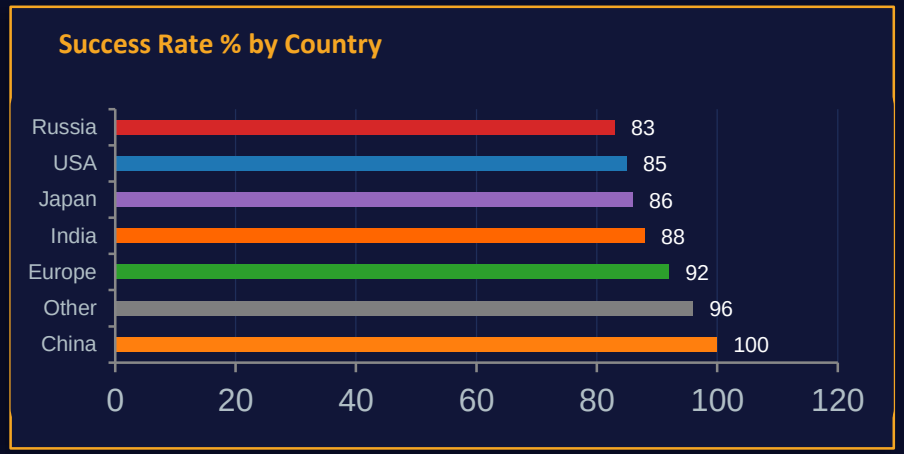
1. Raw CSV (36 cols)
2. Python Cleaning
3. 23-col CSV
4. MySQL Raw Table
5. Type Conversions
6. missions_clean
7. 7 SQL Views
8. **Power BI Dashboard**

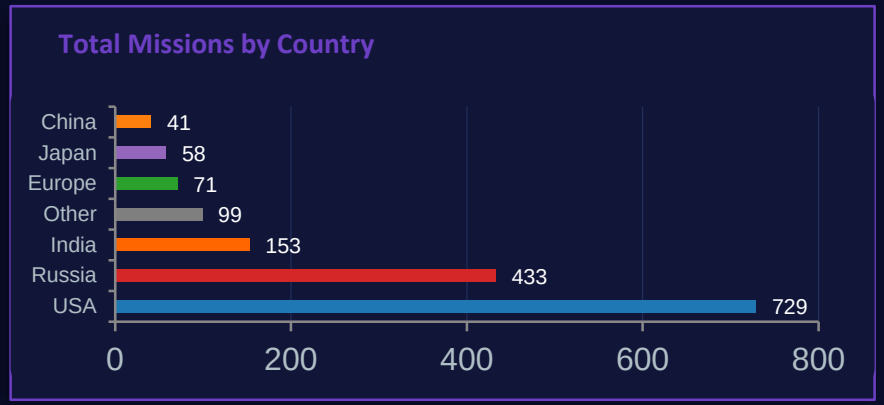
Mission Outcome Split



EXPLORATORY DATA ANALYSIS — KEY INSIGHTS







SQL VIEWS CREATION — 7 VIEWS FOR POWER BI

vw_mission_summary

Core flat table — all 23 key columns. Used by most visuals across all 5 pages.

Filter: launch_year <= 2025, NULLIF(launch_site, '')

vw_orbit_analysis

Orbital distribution — mission count, avg mass, success rate per orbit class.

Filter: orbit_class NOT IN ('Unknown','Other')

vw_country_performance

Country KPIs — total missions, success rate, avg cost, type diversity per country group.

Filter: launch_year <= 2025

vw_launch_site_map

Geographic data — 47 launch sites with coordinates, success rate, first/last launch year.

Filter: launch_site IS NOT NULL AND != ''

vw_yearly_trends

Time series — missions and success rate by year, decade, and country for trend charts.

Filter: launch_year IS NOT NULL AND <= 2025

vw_technology_evolution

Fuel & propulsion trends — missions and avg cost by decade, fuel type, mission type.

Filter: fuel_type != '' AND decade IS NOT NULL

vw_rocket_reliability

Vehicle analysis — reliability %, total launches, payload avg. HAVING total_launches >= 3.

Filter: launch_year <= 2025, min 3 launches

Key Principle — Each view uses the correct WHERE filter based on how that column stores missing data (NULL vs '' vs 'Unknown'). Wrong filter = wrong numbers.

DASHBOARD DEVELOPMENT — 5-PAGE DESIGN

6-Page Report Structure

Home	Navigation hub with 5 dashboard buttons
Page 1	Executive Overview — KPIs + trends
Page 2	History & Trends — decade analysis
Page 3	Country & Agency — deep dive
Page 4	Rockets & Launch Sites
Page 5	Mission & Orbital Analysis

Global Slicers — Synced Across All 5 Pages

country_group · launch_year (range) · decade (tiles) · mission_outcome · mission_type

Enable sync: View → Sync Slicers → tick all 5 pages for each slicer

7 DAX Measures

Total Missions · Success Rate % · Failure Rate %
Avg Mass kg · Avg Cost USD M
Missions With Cost · Cumulative Missions

DASHBOARD 1 — EXECUTIVE OVERVIEW

1,584
Total Missions

85.9%
Success Rate

68 Yrs
Year Span

333
Unique Rockets

Key Insights

USA Leads

729 missions — 46% of all launches. Followed by Russia/USSR with 433 (27%).

Space Race Peak

1960s account for 976 missions — more than all other decades combined.

High Success

1,361 successes vs 221 failures. Overall success rate of 85.9%.

Mission Mix

Military/Recon (26%) and Earth Observation (17%) dominate mission types.

DASHBOARD 2 — HISTORY & TRENDS ANALYSIS

57%
1950s Success Rate

94%
2010s Success Rate

976
1960s Missions

64%
Liquid Fuel Share

Key Insights

Dramatic Improvement

Success rate rose from 57% in the 1950s to 94% in the 2010s — a 37-point gain.

Cold War Spike

1960s Space Race drove an unprecedented volume peak — 62% of all recorded missions.

Fuel Shift

Liquid propellant dominates at 64%. Ion/Electric engines rising sharply from 2010s onward.

Country Heatmap

Russia/USSR was absent post-2010s in the matrix — replaced by China's rapid expansion.

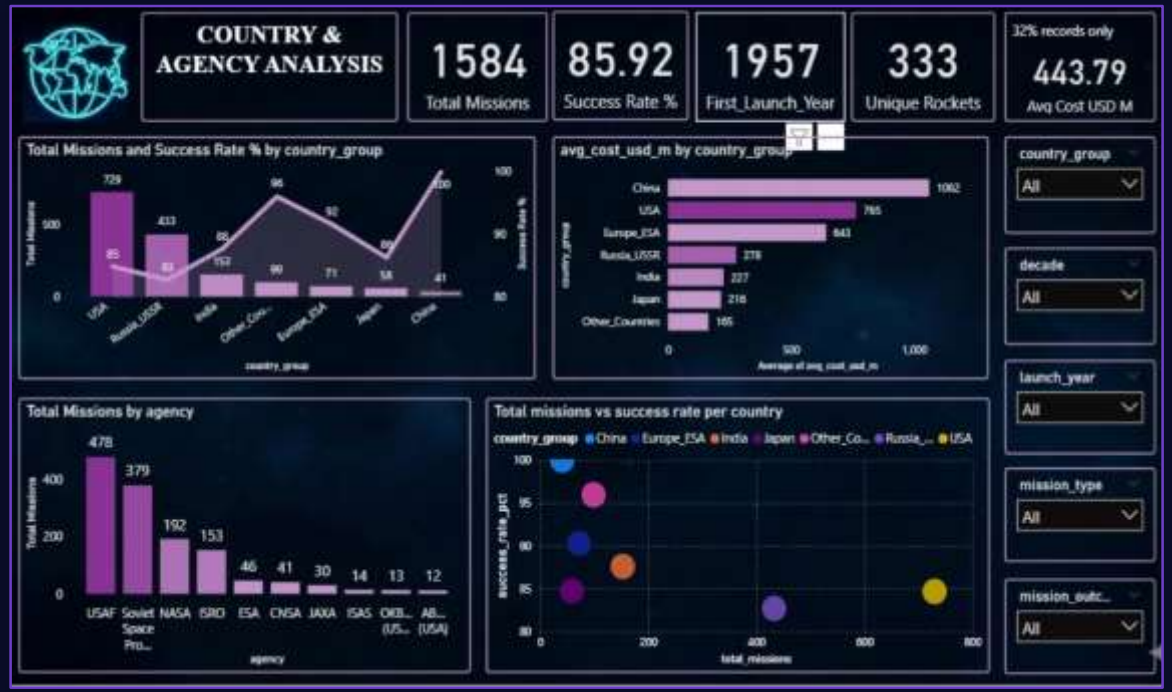
DASHBOARD 3 — COUNTRY & AGENCY ANALYSIS

100%
China Success Rate

$1,062M
China Avg Cost (highest)

$227M
India Avg Cost (lowest)

478
USAF Missions (top agency)



Key Insights

China Perfection

China achieved a 100% success rate across all 41 recorded missions — most remarkable record.

ISRO Rising

India's ISRO shows the steepest modern growth — 88% success on significantly lower budgets.

Cost Leaders

China avg $1,062M is highest; India $227M is lowest — over 4x difference in cost efficiency.

Agency Dominance

USAF (478) and Soviet Space Program (379) collectively account for 54% of all missions.

DASHBOARD 4 — ROCKETS & LAUNCH SITES

82
Voskhod Launches (most used)

318
Vandenberg Launches (top site)

95%+
Delta II / Soyuz-FG Reliability

333
Unique Rocket Variants

Key Insights

🚀 Most Flown

Voskhod leads with 82 launches. Top 5 rockets are all Cold War-era US/Soviet vehicles.

🔑 Top 3 Sites

Vandenberg (318), Baikonur (250) and Cape Canaveral (220) — 3 sites = 50% of all launches.

⚕ High Reliability

Delta II, Soyuz-FG, Vega and H-IIA all exceed 95% reliability — proven veteran platforms.

⚡ Falcon 9 Era

The matrix shows Falcon 9 emergence in 2010s — representing the commercial reusable revolution.

DASHBOARD 5 — MISSION & ORBITAL ANALYSIS

715
LEO Missions (45% of total)

94%
SSO Success Rate (highest)

8,645 kg
Avg Lunar Payload (heaviest)

77%
Mars Success Rate (lowest)

Key Insights

LEO Dominates

Low Earth Orbit hosts 715 missions (45%) — driven by ISS, Earth observation, and cubesats.

SSO Precision

Sun-synchronous orbit has the highest success rate at 94% — well-understood and repeatable.

Lunar Heavy

Lunar missions carry the heaviest average payloads at 8,645 kg due to propulsion demands.

Mars is Hard

Mars/Interplanetary has the lowest success at 77% — deep-space navigation remains the frontier.

INSIGHTS, CONCLUSION & RECOMMENDATIONS

Key Insights

🏆 **Success Rate**	China achieved 100% success rate. Russia/USSR lowest at 83%, reflecting early Cold War failures.
📈 **Space Race Peak**	1960s had 976 missions — 62% of all missions. Decade accounts for biggest spike in launch activity.
🚀 **Most Launched**	Voskhod (82 launches) is the most used rocket. USA and Russia dominate top 10 vehicles.
🌍 **Launch Sites**	Vandenberg (318), Baikonur (250) and Cape Canaveral (220) are the top 3 sites globally.
🛰 **Orbit Preference**	LEO dominates at 715 missions (45%). SSO, Lunar and GEO follow as next most used orbits.
💰 **Cost Data**	Only 32% of missions have cost data. China avg $1,062M is highest, India lowest at $227M.
⚡ **Fuel Evolution**	Liquid fuel dominates (64%). Ion/Electric rising in 2010s-2020s for commercial satellites.

Conclusion

- Successfully cleaned 36-column raw dataset to 23 production-ready columns.
- MySQL import pipeline with proper type handling eliminates all data errors.
- 7 SQL views provide clean, filtered data optimised for each dashboard page.
- 5-page Power BI dashboard delivers actionable insights with global slicer sync.
- 88% of missions are fully verified via Launch Library 2 UUID traceability.

Recommendations & Future Scope

- Fill cost data gaps using public sources to increase 32% coverage.
- Add launch coordinates for all 47 sites to enable accurate map visuals.
- Automate monthly data refresh by connecting to Launch Library 2 API.
- Add crew manifest data to enhance Human Spaceflight mission analysis.
- Build ML model to predict mission success probability from vehicle + orbit.
- Expand to include payload operator and customer for commercial analysis.